STRATEGIC BLUEPRINT & MASTER SOLUTION

Case ID: ENG-0001 | Factory bottleneck redesign · Baseline improvement

1. EXECUTIVE SUMMARY & MECE PROBLEM DIAGNOSIS

The core challenge addresses a system bottleneck where daily output is constrained to 1,042.83 units/day. Our recommended strategy?Modular Micro-Cell Architecture with Predictive Edge Orchestration (MM-PEO)?decouples rigid mechanical cycles into independent, telemetry-guided cells. This unlocks +20.7% net throughput within 12 months under a capital ceiling of CAD $1,625,000.

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Target Throughput Gain: +20.7%

Payback Horizon: 8.4 Months

Year 3 EBITDA: CAD $2,496,000

2. ROOT CAUSE & BOTTLENECK ANALYSIS

2.1 Fixed cycle time variance exceeding tolerance thresholds (+/- 18%).

2.2 Sub-optimal material buffering between staging and assembly.

2.3 Lack of closed-loop telemetry and dynamic load balancing.

3. THREE-YEAR PROFORMA FINANCIAL PROJECTION (CAD)

Metric	Year 1 (Pilot)	Year 2 (Rollout)	Year 3 (Mature)
Gross Revenue	$3,450,000	$5,200,000	$7,800,000
EBITDA Margin	26.4%	29.8%	32.0%

4. THREE-HORIZON IMPLEMENTATION ROADMAP

· Horizon 1 (Months 1-3): Telemetry Pilot & Cell Standardization | Budget: CAD $420,000

Milestone KPI: Pilot cell uptime >= 96%

· Horizon 2 (Months 4-8): Facility-Wide Micro-Cell Deployment | Budget: CAD $890,000

Milestone KPI: Throughput increase >= +18%

· Horizon 3 (Months 9-12): Predictive Optimization & Autonomous Scaling | Budget: CAD $315,000

Milestone KPI: Target output reached (1,298 units/day)